

SYED SULMAN AHMAD

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WORK EXPERIENCE

CAMP-PhyRE Lab – University of Washington

Jan 2024 – Present, Seattle, WA

Projects:

Teaching Assistant ME569 – Data-Driven Control and Machine Learning – supervisor: Dr Steven L Brunton

1. Developed a model predictive attitude controller for a spacecraft that realigns after capturing irregular debris, handling actuator saturation and disturbances, validated it on large pointing error, pure de-tumbling, and combined displacement-rate scenarios.
2. Data-Driven Flight Dynamics & Control: Developed reduced-order nonlinear aircraft models using JSBSim (6-DoF) and SINDy from aggressive maneuver datasets, capturing key aerodynamic couplings in sparse state-space form and integrating them into an MPC framework for real-time, high-agility trajectory tracking, with direct relevance to onboard guidance.
3. Modelled RL-MPC control for Fanuc CRX arm robot, design sim-to-real pipelines where policies are trained in simulation.
4. Supervised multi agent robot coordination projects using Crazyflie class quadrotors and ground robots, focusing on distributed control and RL based policies for formation keeping, obstacle avoidance via (CBF-QP), and collision free navigation in settings.

Research Assistant – US Army Corps of Engineers (ERDC) – PI: Dr John Palmore

5. Turbulent Kinetic Energy (TKE) Budget Closure: Overcoming literature gap in multiphase spray modeling by resolving the full anisotropic Reynolds stress tensor and third-order velocity moments to develop underwater acoustic deterrent systems to stop carp migration in great rivers protect Mississippi river on east coast, delivering operational system to US Army Corp ERDC.
6. Developed underwater acoustic deterrent systems to protect Mississippi River ecosystems by solving linear acoustic wave equations through **finite element methods on HPC cluster**, delivering operational “BASIC” system for US Army ERDC.

Design and CFD Intern (Hokie Flying Club and Formula SAE) – Virginia Tech

Aug 23 – Dec 23, Blacksburg, VA

Projects:

- Formula Student Vehicle for Weight Optimization: Redesigned chassis subcomponents using topology optimization, achieving 10% mass reduction at critical weld joints while maintaining stress constraints (FoS > target), for 5% increase in driver crash protection.
- Designed and optimized the FS car’s rear suspension from first principles—including geometry, A-arm hard-points, and load-transfer behavior using custom MATLAB scripts—while reducing subsystem weight by 6% and validating key components (engine mount, sprocket, powertrain brackets) through FEA for manufacturable SolidWorks/NX designs
- Steering System Design for Formula Prototype: Developed a performance-oriented steering wheel and mounting interface with optimized load paths and ergonomic constraints. Reduced structural mass by 25% while preserving stiffness under steering torque loads; validated using FEA (von Mises stress, deformation limits) and aligned with Formula Student packaging constraints.
- Led the end-to-end design and testing of fixed-wing VTOL and autonomous quadcopter UAVs, including full quadcopter design from concept to prototype (battery-locking mechanisms with GD&T), compact fuselage surfacing for integrated electronics, LiDAR/TOF sensor-equipped 3D-printed prototypes, and wind-tunnel/CFD campaigns in Ansys Fluent and NX/SolidWorks that cut simulation setup time by about 15–20% while meeting strict weight and FOS targets via FEA.

Logistics Officer (Supply Chain) – Velo Group - BAT (British American Tobacco)

Nov 21 – Jan 22, Pakistan

- Oversaw a team of 20 floor laborers, 10 general laborers, 2 supervisors, and worked with foreign and local vendors.
- Led a quality initiative for a new product line, implemented key quality indicators (defect rates, yield, equipment uptime) and validating formulas, equipment, and processes, resulted 10% reduction in defects through staff training and process optimization.
- Kaizen Event for Scrap Reduction: Organized a Kaizen event to address scrap in the molding process, identifying and eliminating sources of waste. Achieved a 35% reduction in scrap rates, saving \$75,000 in material costs annually for Finished Goods unit.
- Supplier Quality Enhancement Program: Led a project to develop and implement a Supplier Quality Scorecard system, providing real-time performance metrics. Improved supplier consistency by 20% through quarterly reviews and corrective action plans.

EDUCATION

PhD (Multiphase Flows and Acoustics) – University of Washington, Seattle

Present - June, 2027

MSc (Mechanical Engineering) – University of Washington, Seattle

Graduated, Dec 2025

Semester Exchange, Sungkyunkwan University, Seoul, South Korea

Aug 2018 – Dec 2018

Bachelor of Mechanical Engineering – NUST, Pakistan

Graduated, Jul 2020

SKILLS

CFD/FEA:	SolidWorks, Fusion 360, NX, ANSYS (Fluent), OpenFOAM, COMSOL Multiphysics 6.3, STAR-CCM+
Programming/Coding & OS:	MATLAB, C/C++, Python, Linux OS, Microsoft Office software, Git, HPC, SLURM, CUDA
ML & Control:	LQR, Kalman Filter, surrogate modeling, DQN, policy gradient, ROM, Non-Convex Optimization